

REVIEW

The Role of Intraoral Scanners in the Shade Matching Process: A Systematic Review

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Abstract

Purpose: The variation in findings with regards to the accuracy and precision of intraoral scanners for shade selection are no doubt confusing for clinicians who may find it difficult to make evidence-based decisions. The aim of this systematic review is to provide a comprehensive and in-depth assessment of available studies to determine the viability of using intraoral scanners for the purpose of shade matching. The PICO-guided research question is as follows: when shade matching, are intraoral scanners as valid as visual or other digital shade measuring devices in determining tooth colors.

Methods: Electronic databases including PubMed/MEDLINE, SCOPUS, EBSCO, Cochrane, and ProQuest were systematically searched for articles published between January 1, 2011 and December 30, 2021 using the main search terms: “intraoral scanners,” “scanners,” “TRIOS,” “CEREC,” “Planmeca,” “Medit,” “digital dentistry” in concurrence with one of the following keywords: “EasyShade” OR “shade selection” OR “shade matching” OR “shade” OR “tooth color” OR “tooth shade” OR “digital shade matching.” Bibliographies of included articles and the following journals were searched for relevant articles: *Journal of Prosthetic Dentistry*, *Journal of Prosthodontics*, *Journal of Esthetic and Restorative Dentistry*, *Journal of Advanced Prosthodontics*, and *Journal of Dentistry*. A total of 15 articles were included in the review.

Results: Intraoral scanners are highly repeatable for shade matching, and outperformed visual shade matching. Accuracy varied significantly between studies, with the majority recommending the use of visual shade matching to confirm/verify the intraoral scanner results. Setting intraoral scanners to the Vita 3D Master shade guide improved both accuracy and precision. Shade matching with intraoral scanners may be influenced by external factors such as ambient light sources and incorrect use or manipulation.

Conclusion: Intraoral scanners set to the Vita 3D Master shade guide may be used for shade matching, but shade should be verified with visual shade matching. Further studies are required to address limitations of current studies.

KEYWORDS

Color science, digital dentistry, intraoral scanners, shade matching, shade selection

Successful shade matching of restorations to the natural dentition is a challenging task. This challenge arises as a result of the complex optical properties of teeth which are laborious to replicate, as well as the highly subjective nature of color.¹ Historically, color was difficult to evaluate, describe, and

reproduce. This changed when Munsell published the Atlas of the Munsell Color System where he described the color as a 3-dimensional phenomenon composed of hue, chroma, and value.^{2,3} To this day, the Munsell color system is used to describe what is known as the primary optical characteristics

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of teeth.⁴ The Vita Classic shade guide is based on this system with the letters representing hue, and the numbers representing chroma.³ Munsell's Color System, however, provides a qualitative measure of color without numerical values.

In 1976, the International Commission of Illumination (Commission Internationale de l'Eclairage; CIE) introduced a new method for describing and quantifying color through what they referred to as the color space.⁵ The CIE $L^*a^*b^*$ color space is based on three coordinates, where L^* represents value or lightness from black to white, and a^* and b^* represent chromatic value on red-green and yellow-blue axes, respectively. The CIELAB coordinates resulted in the development of Delta E (ΔE), a mathematical formula that calculates and quantifies color differences between two measured values. In dentistry, ΔE was used to study and set parameters for perceptibility and acceptability thresholds which define visual match or mismatch of restoration in terms of color.⁶ Fundamentally, perceptibility thresholds determine when a difference is perceived by the observer, whereas acceptability thresholds mark the point after which the mismatch becomes clinically unacceptable. Multiple ΔE values for acceptability and perceptibility have been reported in the literature.⁶

One of the shortcomings of the CIELAB system is the lack of uniformity when small color changes are involved. To address this, the International Commission of Illumination introduced an updated formula known as CIE2000. A new color difference formula, CIEDE2000 (ΔE_{00}), was also introduced. Recent studies have shown that CIEDE2000 outperformed other color difference formulas and allows for better evaluation of visual tolerances.⁶

Today, patients are presenting with high esthetic demands which, in turn, has increased the need for highly esthetic restorations that replicate the optical properties of the natural dentition.⁷ The most common method used to accomplish this task is visual shade selection using shade guides.^{8,9} The visual shade matching process produces highly varying results and is influenced by multiple factors that include light viewing conditions, eye fatigue, observer interpretation, clinical experience, and proficiency.^{9,10} It, therefore, comes as no surprise that numerous studies have reported a high percentage of color mismatch in ceramic crowns.^{11,12}

Consequently, different methods of shade selection have been developed and introduced to replace or augment the visual shade selection method. These include digital cameras, spectrophotometers, spectroradiometers, and colorimeters. Various studies have been conducted to compare these instruments to visual shade matching using quantifiable values to calculate the color difference (ΔE) with many shade matching instruments showing remarkable results.^{13–19}

Recently, intraoral scanners have been updated to utilize their high-definition cameras for the purpose of shade matching. This is advantageous to clinicians as it combines impression making and shade selection in one tool. Different studies have been conducted to assess the accuracy and repeatability of intraoral scanners when used for shade matching. However, findings and conclusions varied significantly. This systematic review aimed to assess and compare

the available literature on intraoral scanners and shade matching in order to help clinicians make evidence-based decisions on their use in practice.

METHODS

The research question for this review, formulated based on the problem, intervention, comparison, and outcome (PICO) guidelines, was as follows: when shade matching, are intraoral scanners as valid as visual or other digital shade measuring devices in determining tooth colors. Two independent reviewers (MA, DM) systematically searched five electronic databases that included PubMed/MEDLINE, SCOPUS, EBSCO, Cochrane, and ProQuest for articles published between January 1, 2011 and December 30, 2021. The main search terms “intraoral scanners,” “scanners,” “TRIOS,” “CEREC,” “Planmeca,” “Medit,” “digital dentistry” were used in concurrence with one of the following keywords: “EasyShade” OR “shade selection” OR “shade matching” OR “shade” OR “tooth color” OR “tooth shade” OR “digital shade matching”. Each database was searched using the different keyword and articles returned were screened for relevance. The electronic database search was complemented by a manual search of bibliographies and a review of grey literature for relevant articles that met the inclusion criteria. To minimize the risk of omitting relevant sources, a systematic approach to reviewing grey literature based on four different strategies that included grey literature databases (i.e., NGO Search, OpenGrey), customized Google search engines, targeted websites, and consultation with contact experts was followed.²⁰ Furthermore, the following journals were hand searched for related studies published between January 1, 2011 and December 30, 2021: *Journal of Prosthetic Dentistry*, *Journal of Prosthodontics*, *Journal of Esthetic and Restorative Dentistry*, *Journal of Advanced Prosthodontics* and *Journal of Dentistry*.

All titles and abstracts were first assessed for the following inclusion criteria which included clinical or in vitro studies published in PubMed indexed peer-reviewed journals that evaluated or compared the accuracy or precision of intraoral scanners in the shade matching process to visual or other digital shade matching instruments. Studies that involved intraoral scanners without a focus on shade matching and those that assessed shade matching of soft tissue or maxillofacial structures were excluded. Moreover, articles published in nonpeer reviewed journals, abstracts lacking full text, book chapters, case reports, and studies published in languages other than English were also excluded. Two calibrated reviewers (MA, DM) extracted the data from eligible articles into structured tables. Both authors reviewed the articles independently before comparing and discussing results. Studies with dissimilarities in findings were re-reviewed by a third reviewer (FZ).

Included articles were assessed for methodological quality and risk of bias using the Joanna Briggs Institute Critical Appraisal Checklist for Quasi-Experimental Studies (Table 1).²¹ Two reviewers (MA, DM) independently assessed the included studies before discussing appraisal

TABLE 1 The Joanna Briggs Institute Critical Appraisal Checklist for Quasi-Experimental Studies

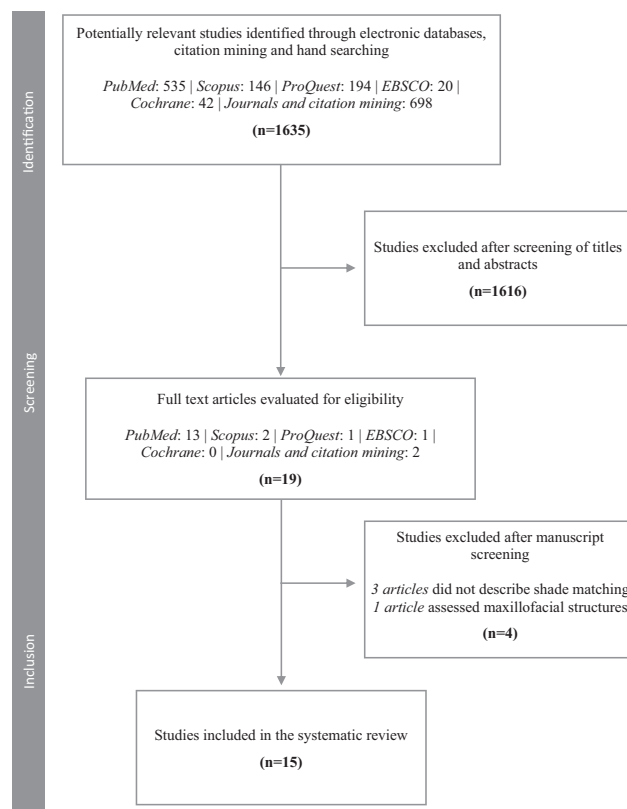
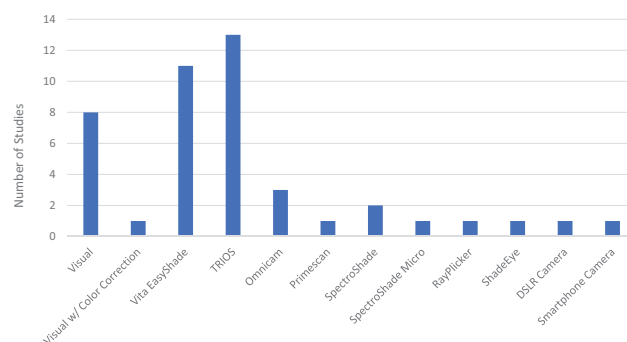
Question	Answer
1 Is it clear in the study what is the “cause” and what is the “effect” (i.e., there is no confusion about which variable comes first)?	Yes, no, unclear or not applicable
2 Were the participants included in any similar comparisons	
3 Were the participants included in any comparisons receiving similar treatment/care other than the exposure or intervention of interest?	
4 Was there a control group?	
5 Were there multiple measurements of the outcome both before and after intervention/exposure?	
6 Was follow-up complete and if not, were differences between groups in terms of their follow-up adequately described and analyzed?	
7 Were the outcomes of participants included in any comparisons measured in the same way?	
8 Were outcomes measured in a reliable way?	
9 Was appropriate statistical analysis used?	

outcomes. A third reviewer (FZ) re-assessed studies with a lack of consensus. No articles were excluded based on the outcome of the critical appraisal.

RESULTS

The search strategies yielded a total of 1635 potential articles. After evaluation of titles and abstracts, 19 full-text articles were identified and reviewed. Four articles were further excluded as they did not meet the inclusion criteria, and 15 studies were included in the final review (Fig 1). The JBI Critical Appraisal Checklist for Quasi-Experimental Studies showed a low overall risk of bias for all included articles. After the agreement between the three reviewers, question 4, which assessed the presence or absence of a control group, was evaluated based on the use of a control device or instrument in the study. Outcome reliability was considered to be high risk in studies where conversion tables were used prior to comparing instruments. Questions 3, 5, and 6 were not applicable for any of the included studies. Table 2 provides an overview of the appraisal outcome and risk of bias.

The studies included in this review assessed only three intraoral scanners: TRIOS 3 (3Shape; Copenhagen, Denmark),^{22–34} CEREC Omnicam,^{23,30,35} and CEREC Primescan (Dentsply Sirona; York, PA).²³ The TRIOS 3 was the most commonly evaluated scanner.^{22–34} Reported results varied between studies, as did the methodology and devices used for comparison. Eleven out of the 15 studies used one of the Vita EasyShade spectrophotometers (EasyShade, EasyShade V, EasyShade Advance; VITA; Bad Säckingen, Germany)^{22,23,25,27–30,32–35} as a reference

**FIGURE 1** Flowchart outlining the study identification, screening, and inclusion process.**FIGURE 2** The different methods of shade selection used to assess precision and accuracy of intraoral scanners.

device. Other reference devices included spectrophotometers and colorimeters such as SpectroShade (MHT Optic Research AG; Zürich, Switzerland),^{24,29} ShadeEye-NCC (Shofu Inc., Kyoto, Japan),³¹ and RayPickler Quickshade (Borea Dental; Limoges, France),³⁰ as well as smartphone and DSLR cameras.³² Most studies^{22,26–30,33,34} included visual shade matching in their comparison to different instruments that included intraoral scanners. One study, however, compared intraoral scanners solely to visual shade matching (Fig 2).²⁶

TABLE 2 Summary of the quality assessment and critical appraisal using the Joanna Briggs Institute Critical Appraisal Checklist for Quasi-Experimental Studies. Included studies showed a low overall risk of bias

Study	Q1	Q2	Q4	Q7	Q8	Q9	Appraisal Outcome
Liberato et al ²²	⊕	⊕	⊕	⊕	⊕	⊕	⊕
Ebeid et al ²³	⊕	—	⊕	⊕	⊕	⊕	⊕
Rutkūnas et al ²⁴	⊕	⊕	⊕	⊕	⊖	⊕	⊕
Revilla-León et al ²⁵	⊕	⊕	⊕	⊕	⊕	⊕	⊕
Reyes et al ²⁶	⊕	⊕	⊕	⊕	⊕	⊕	⊕
Czigola et al ²⁷	⊕	⊕	⊕	⊕	⊕	⊕	⊕
Brandt et al ²⁸	⊕	⊕	⊕	⊕	⊖	⊕	⊕
Mehl et al ²⁹	⊕	⊕	⊕	⊕	⊖	⊕	⊕
Hampé-Kautz et al ³⁰	⊕	⊕	⊕	⊕	⊖	⊕	⊕
Yoon et al ³¹	⊕	—	⊕	⊕	⊖	⊕	⊕
Sirintawat et al ³²	⊕	—	⊕	⊕	⊖	⊕	⊕
Fattouh et al ³³	⊕	⊕	⊕	⊕	⊕	⊕	⊕
Culic et al ³⁴	⊕	⊕	⊕	⊕	⊕	⊕	⊕

⊕ = Low risk ⊖ = High risk — = N/A

Precision

Precision, reliability, or repeatability is a measure of consistency. For color measuring instruments, precision reflects their ability to reproduce the same results in measurements that are made under the same conditions as the original measurement.^{23,26} Various studies assessed the precision of intraoral scanners in the shade matching process. Liberato et al²² found that the TRIOS 3 scanner configured for the Vita 3D Master shade guide showed the highest repeatability when compared to the Vita EasyShade spectrophotometer and visual shade matching under corrected light. All other studies^{24,26–29,31–33} that assessed repeatability of intraoral scanners in comparison to visual shade matching or the Vita EasyShade reported similar findings with the scanner showing similar or superior repeatability to both methods (Fig 3). In the study by Ebeid et al,²³ results showed that although all three studied scanners (TRIOS 3, CEREC Omnicam and CEREC Primescan) were superior to the Vita EasyShade, all four devices showed poor repeatability (<52%). This was the sole study that reported unsatisfactory repeatability of intraoral scanners.

The TRIOS 3 scanner showed no significant differences in reliability when compared to the ShadeEye-NCC colorimeter, and both instruments showed excellent repeatability.³¹ In addition, Rutkūnas et al reported that the TRIOS 3 scanner had similar repeatability to the SpectroShade spectrophotometer.²⁴ Although both Vita Classic and 3D Master shade guide settings resulted in good overall repeatability (>87%), repeatability improved when the scanner was set to record color using the Vita 3D Master shade guide. A literature review by Tabatabaian et al³⁶ on different methods for tooth shade selection confirmed the overall high precision of intraoral scanners (>85%).

Accuracy

Accuracy reflects how close measurements are to their true value. For color measuring instruments, accuracy is determined by comparing provided readings against a gold standard instrument or reference value.²³ The accuracy with which intraoral scanners measured tooth shade varied significantly between studies (Table 3). Accuracy generally decreased when the scanner was set to report Vita Classic shades, akin to what was reported with precision. Different studies reported different accuracy values for the TRIOS 3 scanner which included 43.9%,²⁸ 53.3%,²⁴ and 66%.²³ Ebeid et al²³ concluded that the accuracy of the CEREC Omnicam and Primescan was 57% and 63%, respectively, and that caution must be exercised when using intraoral scanners for shade matching due to low overall accuracy (<80%). The correlation between the CEREC Omnicam and the Vita EasyShade Advance was found to be poor (<23%).³⁵ As a result, the Omnicam should not be used as a solo tool for shade matching. A clinically significant difference was also noted in ΔE values between the Vita EasyShade V, the reference device, and the CEREC Omnicam and TRIOS 3 scanners.³⁰ When comparing TRIOS 3 to the ShadeEye-NCC colorimeter, L*A*B* values differed significantly.³¹ Interestingly, L* and B* values strongly correlated, but the difference in overall coordinates meant that scanner accuracy cannot be guaranteed.

One study showed that scanners had a tendency to calculate shades with a lower value.²⁵ This was contradicted by other studies that noted shades with a higher overall value when intraoral scanners were used.^{28,30,36} This higher value was attributed to the unpolarized light source produced by the scanners.^{30,36} Furthermore, ambient light conditions had a significant impact on the accuracy of intraoral scan-

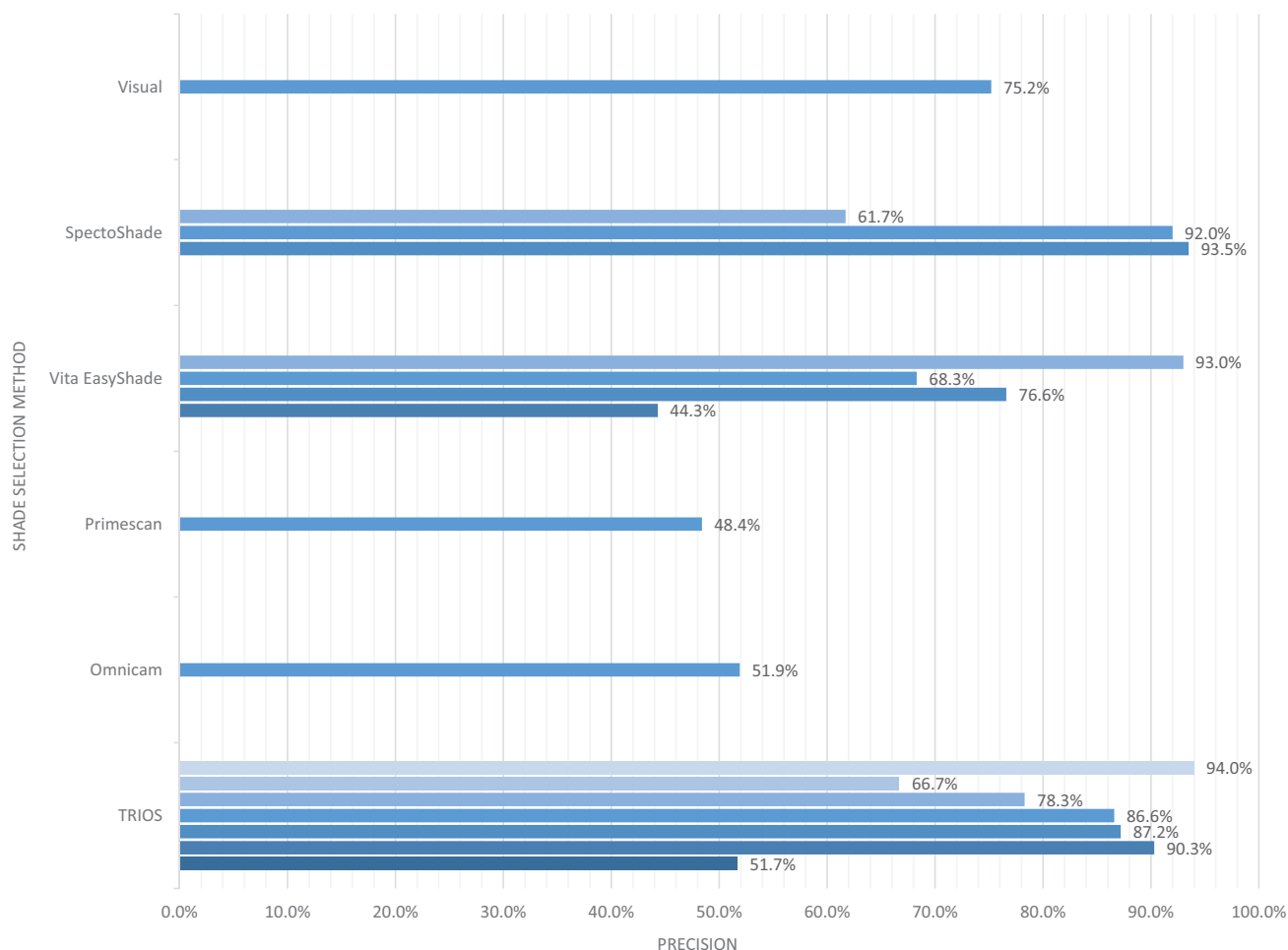


FIGURE 3 Precision of different methods of shade selection in articles reporting percentages. Instruments, including intraoral scanners, showed significantly higher precision when compared to visual shade matching.

ners but did not affect spectrophotometers such as the Vita EasyShade.²⁵

Only two studies concluded that intraoral scanners outperform all other shade selection methods in terms of accuracy.^{22,32} A third study found that the TRIOS 3 could be used as an alternative to the Vita EasyShade as they both provided shade values with no statistically significant differences.³⁴ Mehl et al²⁹ observed that although there was a statistically significant difference in shade selection between the TRIOS scanner and multiple spectrophotometers (SpectroShade, SpectroShade Micro, EasyShade and EasyShade Advance), the results were of no clinical significance when assessing ΔE values ($\Delta E < 3.3$). Intraoral scanners could therefore be used for accurate shade matching.

It is noteworthy to mention that out of the 12 studies that looked at accuracy, five^{23,24,27,28,31} concluded that intraoral scanners should be supplemented with additional methods to determine the appropriate shade, one study recommended caution with their use,²³ and 2 studies^{30,35} concluded that scanners should not be used for shade selection.

Influencing factors

Numerous factors appeared to have a direct influence on the shade matching capabilities of intraoral scanners and, as a result, impacted the findings of some studies. The choice of shade guide when setting up the scanner for shade matching was one of the most commonly reported influencing factor,^{22,24,25,27,35} with the Vita 3D Master settings resulting in improved accuracy in most studies.

Revilla-León et al²⁵ studied the effect of ambient light conditions, or room illuminance, on the shade matching capabilities of intraoral scanners and assessed the performance of the TRIOS 3 scanner under different color lux (lx) illumination ranging from 0 to 10,000 lx. Lux is measure of illuminance and refers to the total amount or intensity of the light that falls onto a surface. It is different from lumens which describes luminous flow, or the total amount of light emitted by a source in all directions. They concluded that ambient light sources had a significant impact on the shade matching capabilities of intraoral scanners with no light (0 lx) providing the highest consistency when compared to the

TABLE 3 Accuracy of the different methods of shade selection with sample size, study type and potential confounding variables identified. Accuracy showed high variation between studies

Study	Sample size	Study type	Variables	Outcome
Liberato et al ²²	28 teeth	In vivo	Vita Classic and 3D Master shade guides used.	Shade matching instruments were more accurate than visual shade matching. TRIOS scanner set to Vita 3D Master was most accurate.
Ebeid et al ²³	10 blocks	In vitro	Monochromatic blocks measured.	No statistically significant difference between Vita EasyShade (78%) and TRIOS (66%). Significant differences seen with Primescan (63%) and Omnicam (57%). Low overall accuracy (<80%).
Rutkūnas et al ²⁴	120 teeth	In vivo	Vita Classic and 3D Master shade guides used. Shades converted to CIELAB values using conversion table.	With SpectroShade as a reference, accuracy of the TRIOS scanner was 53.3% when set Vita 3D Master and 27.5% when set to Vita Classic.
Revilla-León et al ²⁵	3 teeth	In vivo	Shade matching at different color lux (0, 500, 1,000 and 10,000 lx)	TRIOS provided a lower shade than Vita EasyShade, and performance was influenced by ambient light conditions (best performance at 0 lx).
Czigola et al ²⁷	10 teeth	In vivo	Vita Classic and 3D Master shade guides used.	Visual shade matching using the Vita 3D Master shade guide was most accurate (34%), followed by Vita EasyShade (27%), TRIOS (22%) and Vita Classic (17%).
Brandt et al ²⁸	107 teeth	In vivo	Vita 3D Master shade guide used. Shade converted to CIELAB values using conversion table.	With the Vita EasyShade as a reference, TRIOS accuracy was higher (43.9%) compared to the dentist (35.5%) and lab technician (33.6%).
Mehl et al ²⁹	40 teeth	In vivo	CIELAB values and Vita 3D Master shades recorded.	SpectroShade Micro showed the highest accuracy, followed by SpectroShade. Clinical significance ($\Delta E < 3.3$) showed no difference between all other methods including the TRIOS scanner.
Hampé-Kautz et al ³⁰	40 teeth	In vivo	Vita 3D Master shades converted to CIELAB values using conversion table.	RayPicker produced the smallest ΔE mean value (1.85 ± 0.26) when compared to the reference device. Highest differences were for the Omnicam (2.75 ± 0.23) and TRIOS (3.05 ± 0.27).
Yoon et al ³¹	5 shade tabs	In vitro	Vita Classic shade guide used. Shades converted to CIELAB values using software.	L* and B* values strongly correlated between both devices, but overall coordinates showed significant differences.
Sirintawat et al ³²	30 milled crowns	In vitro	Vita 3D Master shades converted to CIELAB values using conversion table.	Compared to different cameras and spectrophotometers, TRIOS was the only device that provided a shade match with ΔE below the clinical accepted value of 6.8 (ΔE : 5.96).
Yılmaz et al ³⁴	5 teeth	In vivo	Shade matching at different color temperatures (4000, 6500K).	Both the Vita EasyShade and TRIOS scanner showed similar results and were not affected by color temperature.
Culic et al ³⁵	80 teeth	In vivo	Vita Classic and 3D Master shade guides used.	Overall average correlation between Omnicam and Vita EasyShade was 15%, with slightly higher correlation with Vita Classic (17.5%) when compared to 3D Master (12.9%).

Vita EasyShade reference device. In contrast to other studies in this review,^{22,24,36} more inconsistencies were seen amongst different ambient light sources when the Vita 3D Master settings were used when compared to the Vita Classic settings.

A second study assessed the impact of ambient light conditions by shade matching under two different color temperatures (4000 and 6500K). By definition, color temperature is the temperature at which a blackbody that absorbs all frequencies of light would emit radiation of the same color as the light source. The authors found no difference in performance of the intraoral scanner or Vita EasyShade under the different light temperatures.³⁴

Inconsistencies in accuracy and precision between intraoral scanners, colorimeters and spectrophotometers may stem from different factors.³⁶ The size of the analyzed area differs significantly when comparing intraoral scanners to other color measuring devices. When analyzing shade, intraoral scanners examine the entire tooth surface from multiple angles, whereas spectrophotometers and colorimeters sample a small area where the tip of the device comes into close contact with the tooth surface. While shade matching with intraoral scanners is not influenced or affected by the operator, gender, or overall experience, results from all shade matching instruments may be affected by incorrect use or manipulation.³⁶

DISCUSSION

The primary purpose of intraoral scanners is the 3D mapping of intraoral structures to create digital impressions. These devices are designed with special optical configurations and working principles where a light source is emitted to focus on the target object and an image sensor receives back the beam reflected by this object. The reflected beam is then focused on the sensor of the acquisition camera by different focusing systems. As a result, the intensity of the light detected by the camera provides a direct measure of the distance between the scanner head and the object.³⁷ Different light sources, acquisition cameras and working systems such as active triangulation, active wavelength sampling and confocal microscopy have been described by different manufacturers.³⁷ As a result of this variation in light source, camera size and operating mechanism, intraoral scanners vary in terms of accuracy and precision with digital impressions.^{38–40} Therefore, similar variations with shade matching are expected.

In contrast to intraoral scanners which have to perform multiple complex algorithms to generate digital impressions as well as detect shade, spectrophotometers and colorimeters are specialized devices that are designed for the purpose of color detection. This is reflected in the internal and external design such as the probe/tip size and angle, lighting source, sensors, filters, and analyzing software which calculates the final shade.³⁵ Despite these features that were designed specifically for shade matching, the reviewed articles showed significant variation in performance between different spectrophotometers and colorimeters, with some studies showing better shade matching capabilities of the intraoral scanner.^{22,32} These differences may be attributed to the varying light sources used such as tungsten-filament bulbs, LED lamps or halogen bulbs. The light source affects the way that the tooth is illuminated and how light is reflected back.³⁵ Furthermore factors such as the internal design and viewing geometry, which affect how the instrument compiles and measures reflectance off a tooth, as well as differences in the internal CPU and data processing algorithms have also been associated with the varying outcomes of shade matching instruments.⁴¹

One of the interesting findings was that 11 out of the 15 included articles used a version of the Vita EasyShade spectrophotometers as a reference device. Multiple studies have been conducted to assess the accuracy and precision of the Vita EasyShade with inconsistent findings.^{41–44} These contradictory results and the inability to validate the definitive measuring accuracy brings doubt as to whether Vita EasyShade is suitable for use as the gold standard reference instrument.^{28,44} Furthermore, the comparison of CIELAB readings by the reference device to Vita shades provided by the scanner to determine accuracy is a limitation. Over 40% of the included studies converted Vita Classic and 3D Master readings provided by the intraoral scanner to CIELAB values using a software or conversion charts. The charts used were

inconsistent between studies, with different conversion charts being referenced by different authors.^{24,28,30–33} Inability to objectively compare values provided by the instruments being studied and the need for such conversions brings to question the accuracy of the comparison and validity of subsequent results.

Results from studies that used monolithic blocks^{23,32} to fabricate specimens in this review may not be of clinical relevance as they do not simulate natural teeth which are polychromatic and multilayered with intricate optical properties. The lack of a standardized ΔE value between the studies on what is clinically acceptable further complicates this comparison. The ΔE values above which the results were clinically significant included 2.45,³⁰ 3.3,²⁹ and went as high as 6.8.^{28,32} Further, no studies were found that used the CIEDE2000 (ΔE_{00}) formula. The dependability of physical shade guides as a determinant of accuracy is controversial as shade guides lack standardization in production and may present with color differences even when produced by the same manufacturer.⁴⁵

The findings of this systematic review show that intraoral scanners have high precision with varying accuracy. However, multiple limitations were identified, and overall study designs varied significantly. Well-designed studies with standardized color patches and a gold standard reference instrument are critical to overcome current in vitro study limitations. Use of the new ΔE_{00} formula and standardizing the parameters for perceptibility and acceptability thresholds in future studies is also essential to provide clinicians with findings that are of clinical relevance.

CONCLUSION

Intraoral scanners are evolving, and their use in shade matching may provide better esthetic outcomes, especially if they are supplementing visual shade matching. However, care must be taken to ensure that potential influencing factors are eliminated to maximize accuracy. Within the limitation of this study, a few conclusions can be drawn. Intraoral scanners show high precision when used for shade selection. Evidence on accuracy of intraoral scanners is inconclusive. Using intraoral scanners with the Vita 3D Master shade guide setting improves both accuracy and precision. Ambient light sources may impact the accuracy and precision of intraoral scanners. Visually confirming the shade calculated by the intraoral scanner appears to provide the best clinical outcome.

CONFLICT OF INTEREST

The authors do not have any conflicts of interest in regards to the current study.

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